

Providing a formal system in which philosophical questions could be asked and answered — a key difficulty on one path to safe AGI

July 2024

Abstract

The fact that we can state mathematical problems and arguments formally plausibly gets us a significant fraction of the way along one path to safely getting a great deal of superhuman mathematical work out of an artificial intelligence. If we could formalize philosophical problems and philosophical arguments, we'd likewise be plausibly able to traverse a significant fraction of a path to safely getting philosophical work out of an artificial intelligence, which in turn would plausibly open up a path to a good future for humanity.¹ This paper explores what it would take for a suitable formalization of philosophy to be provided — whether a formalization of philosophy analogous to the one in mathematics would be that helpful after all, what such a formalization would (need to) look like, and what the [obstacles to]/[difficulties with] providing such a formalization are.

1 Introduction

Since about 1925, we've possessed a way to clearly state mathematical problems and a precise notion of what it is to provide a solution to a particular mathematical problem. For example, the following is the twin prime conjecture written in the language of set theory (meta-remark: idk if this example is the best one or if an example is necessary. it will look fucked if we have to translate to base ZFC. perhaps an annotated version would make sense, ie where we eg point out which part just defines the naturals or whatever):

$$\forall n \in \mathbb{N}, \exists p > n \dots$$

A proof of the twin prime conjecture corresponds to a sequence of strings in this same language such that each string is either an axiom or is produced by the application of some inference rule to some strings which appear earlier than it in the sequence. Importantly, as it is decidable whether a given string is an axiom and decidable whether a given string can be produced by some inference rule from some strings from a given finite sequence of strings, it is also decidable whether a given sequence of strings constitutes a proof of a given statement. (Indeed, software exists for checking proofs written in the language of set theory: https://en.wikipedia.org/wiki/Mizar_system.) This makes it look plausible that, conditional on us figuring out how to train an artificial system which is better than humans at general reasoning, we could

2

discuss why having a formal language is helpful for safely getting superhuman mathematical work out of an artificial system. give more examples of such work: the problem of providing an algorithm to solve some well-defined problem, the problem of constructing an efficient data structure which supports some specific queries, the problem of providing a causal model which predicts certain data well, various other mathematical optimization problems

then say that it'd be great if we could do the same for philosophical questions — it would open up a path to safe ASI. say the point of this essay is to understand more clearly what doing this for philosophy would look like + whether it would be possible for philosophy + why it would be harder for philosophy than for mathematics

motivate this by saying that (1) mathematics and philosophy both sorta seem a priori, (2) historically, both used to be done informally, and people sorta set out to formalize them together (Frege), but then we succeeded at formalizing mathematics

1. You might wonder: "Instead of asking this just about philosophy, wouldn't it make more sense to ask for a system in which all our claims and arguments — philosophical, scientific, mathematical, and whatever else there is — could be formalized, since philosophical problem-setting and argumentation is essentially as broad as all good problem-setting and argumentation anyway?" My short reply to this is: "O wise reader, I agree that this is ultimately a more natural question, and indeed, I will argue for this position later in the paper." That said, one (surface) reason for it being natural to seek an analogy between mathematics and specifically philosophy here is that each is often considered to be (largely) a priori (<https://youtu.be/Lrj4E-hAwZc>), whereas science is not. Another reason is that some points we will make about inquiry and argumentation being purposive, reflective, and otherwise importantly dependent on the inquirer are plausibly more forceful when talking about philosophy than when also talking about science (and perhaps particularly unconvincing when talking about mathematics), even though I think they ultimately also importantly apply to science (and even mathematics). (But really, the main reason I'm framing this piece as being about philosophy is that I started writing it for an essay contest that sought essays on AI doing philosophy.)

2. The original version of this paper also contained a formal proof of the twin prime conjecture, but it was cut during editing.

but haven't yet succeeded at formalizing philosophy, (3) people xyz hope to do something like this (Russell (2022), Tegmark and Omohundro (2023), Dalrymple et al. (2024))

address the objection that well obviously philosophy is harder to formalize because philosophy just consists of the problems that we are still struggling to make precise, that contain terms we're still confused about, by saying that ok, even if that's precisely what philosophy is, we are asking instead [what it would look like] / [whether it is feasible] / [why it would be hard] to formalize the following particular example problems (or, more precisely, which common properties of these problems make them harder to formalize than mathematical problems): * list of examples *

meta-remark: actually, i think i'm a bit unhappy with the paragraph above. we could make the rest of the paper such that it makes sense, but i don't think it fits currently. the issue is that we decide to go with discussing the difficulties with constructing a language in which such questions could generally be formalized later, not with what it'd take to formalize some more particular questions. i've put a slightly modified version below that i think makes more sense given the current plan for the rest of the paper, but i've not erased the one above because i haven't thought about this change carefully yet

address the objection that well obviously philosophy is harder to formalize because philosophy just consists of the problems that we are still struggling to make precise, that contain terms we're still confused about, by saying that ok, we are instead asking [what it would look like] / [whether it is feasible] / [why it would be hard] to create a formal system in which the following example problems can be stated and answered: * list of examples * if the construction of a formal system subsuming in which these problems can be discussed would take them outside of philosophy in your view, then you should think of us not as trying to turn philosophy into something that could be carried out formally, but to replace it with something that could be carried out formally

2 To what extent is having a formal system for mathematical proving even helpful for safely getting superhuman mathematical work out of an artificial system?

More precisely, we could perhaps get an artificial intelligence to

1. provide a formal proof or disproof of some mathematical claim we state formally;
2. design an algorithm to efficiently perform some formally specified task;
3. provide a decently-high-objective-function-value solution to some mathematical optimization problem.

Each of these would constitute helpful mathematical work, because (respectively):

1. having a way to cheaply tell that some non-trivial mathematical hypotheses are true/false seems plausibly helpful for mathematical research;
2. one could use the algorithms the AI provides to do the formally specified tasks which we asked for algorithms for;
3. one could use the mathematical optimization problem solutions for the applications in which those optimization problems arose.

Furthermore, it seems plausible that creating such a system and using it this way could be safe for us, due to a combination of the math-AI maybe not 'caring much about anything in the real world' (at least until a capability level higher than that at which it would start to do these kinds of useful mathematical work) (later edit: i now think this is wrong) and due to it being difficult to push the real world in a bad direction much through these limited output channels. (This is not to say that it is fine to publish insightful research on AI theorem-provers — in my opinion, publishing such research is omniscidal.)

That said, let me quickly temper possible excitement about this kind of mathematical work, though. For example, finding an efficient algorithm for a crisp mathematical problem is only a very small fraction of what it takes to create almost any useful piece of software. For a longer example, note that having such an AI prove/disprove the Riemann hypothesis probably by default provides only an ϵ fraction of the expected mathematical value (to human mathematics) that having a human prove the Riemann hypothesis would provide, because the human's proof would probably come with/from an associated novel body of mathematical understanding that is accessible to human mathematicians and [fruitfully relates to]/[can be fruitfully connected to]/[yields insights about] many mathematical matters, whereas our AI's formal proof would by default not provide any of this value. If we could recover [from the formal proof produced by an alien system] [the mathematical concepts employed in it]/[the mathematical concepts the system must have invented/discovered and used to provide it], then we could get a lot more mathematical value out of the proof, but this is potentially very difficult. It is perhaps analogous to the problem of figuring out what an algorithm (for example, https://en.wikipedia.org/wiki/Dijkstra%27s_algorithm, except of course imagine a case where this is an unknown and potentially very alien algorithm) is doing and how it works (like, understanding enough about it to write the main sections of https://en.wikipedia.org/wiki/Dijkstra%27s_algorithm) given a boolean circuit 'implementing that algorithm' (see here, <https://cs.stackexchange.com/a/124062>, for a description of a

standard procedure for going from a Turing machine to a boolean circuit that captures its operation (on a subset of inputs)). Also, if instead of a verifier program simply checking the proof, we start to inspect the proof ourselves, then we should become somewhat more concerned about the system pushing the world in a bad direction. The extent to which we should be worried might depend on further details about how the prover-AI is created. It is possible that we ought to already be concerned about malignity even if proofs are never inspected, as the system might still decide whether to provide a proof or not (in cases where a proof exists) according to which option would better further some malign end, but this worry doesn't seem sufficiently critical to us to analyze it further at the moment.

2.1 Would such a mathematical prover come to control everything despite its limited apparent domain?

3 Clarifying what we'd want from a formal philosophical system

might want to make the problem clearer, eg is this formalization thing something that happens after things fall out of philosophy by becoming sufficiently clear, or are we trying to formalize this process of making things precise itself, or are we doing some other thing?

let's say we're going with the ambitious goal of formalizing general philosophy here, not particular slices; re the list of questions above, we're not asking why it is that any particular question does not have a formalization, but why it is that we cannot ask these questions formally. for example, there is a reasonable sense in which

We have a clear understanding of what it is to ask for the quickest way for the first player to win in Hex (assuming the other player cooperates)? We have a decent understanding of what it is to ask for the quickest way to win in Gex? But we lack a clear grasp of what it is to ask for the quickest way to win in, uhm, TeX.³

distinction: note that when proving something in mathematics, most reasoning doesn't proceed like the eventual proof, and in fact it seems plausible that most reasoning wouldn't faithfully fit in the formal systems known at present at all.⁴ consider: playing around with some examples to 'get a sense' for some particular kind of object, or thinking about which kinds of objects one would need to play around with to get a better sense for what makes a particular problem difficult, or thinking about the high-level similarities between two arguments to encapsulate the common thread in a lemma, or etc (meta: can give many more examples here — i think most mathematical reasoning cannot be formalized in existing systems). still, for math, this is fine for our purposes of getting mathematical work

now make this same distinction for philosophy: try to state the problem of having a language in which philosophical problems and arguments can be stated, and make it clear that this could be helpful even if captures only a small fraction of the reasoning that goes into producing philosophy

Even having said all this about what we're after, unfortunately, the task of formalizing philosophy probably still remains meaningfully unclear. Some pressure in this direction could be exerted by tasks of this kind being very difficult — if this is the case, whenever we conceive of a precise version of this task, we are prone to recognize its difficulty and to seek a different, more tractable, version. At this point, we are unfortunately unable to further mitigate the worry that the task we are setting out to examine remains insufficiently precise, except by saying this:

- Ultimately, we would like to have a formal system in which philosophical questions could be asked and answered which would allow us to be content with outsourcing arriving at philosophical truths to an AI that produces

With the AI being brought into the picture, let us make one last clarification about how the formal system we seek relates to such an AI. While this AI is supposed to provide arguments which live in a particular formal system to settle claims which also live in that formal system, this does not (at least not obviously) imply that the AI itself needs to reason in that same formal system. For counterexamples, imagine, say, systems like the following:

- an AlphaGo-like theorem-prover AI that involves components which are massive neural nets — at least *prima facie*, we can imagine such a system producing formal arguments while not itself reasoning in that formal system any more than AlphaGo 'reasons in chess';
- a system involving a language model which first searches for a proof in natural language and, once such a proof is found, attempts a translation of this proof into the formal language.

3. Though see <https://texnique.xyz/>.

4. Take it from George Pólya Po?lya 1968: "Mathematics is regarded as a demonstrative science. Yet this is only one of its aspects. Finished mathematics presented in a finished form appears as purely demonstrative, consisting of proofs only. Yet mathematics in the making resembles any other human knowledge in the making. You have to guess a mathematical theorem before you prove it; you have to guess the idea of the proof before you carry through the details. You have to combine observations and follow analogies; you have to try and try again. The result of the mathematician's creative work is demonstrative reasoning, a proof; but the proof is discovered by plausible reasoning, by guessing. If the learning of mathematics reflects to any degree the invention of mathematics, it must have a place for guessing, for plausible inference."

Now, we also do not rule out AIs which do reason in that same formal philosophical language (yes, it is unclear to us as well what this means precisely). Indeed, there could turn out to be reasons why a system which capably produces formal philosophical arguments would really need to effectively be reasoning in, say, an extension of that formal philosophical language. Anyway, we make this final clarification because it implies that even if you think the capable provision of mathematical of philosophical arguments would almost certainly require a certain kind of reasoning, this does not place the requirement that such reasoning should be supported on the formal language we are seeking (at least absent further justification), as the thinking that leads to the provision of a philosophical argument in our formal language (at least *prima facie*) needn't happen in this language itself. This is analogous to how it could well happen that noticing some informal analogy to another mathematical context is a crucial element of the thinking that leads a mathematician to solve an open problem while the proof they then provide does not contain any formal correspondent to the thinking-step of making that analogy — in fact, while we cannot make this precise, it seems intuitively plausible that a majority of the reasoning that leads to a mathematical proof could not be formalized given humanity's current understanding of how to formalize mathematical arguments.

4 What are the obstacles to carrying out this formalization?

We now discuss various reasons which together point at, and make sense of, it being difficult to formalize philosophy (in the sense described in the previous paragraph). These reasons serve to give us a better sense of whether the task of formalizing philosophy, but also to identify obstacles to be overcome if a suitable formalization is to be provided. This is particularly the case since each reason we provide can be seen as providing a property that the formal system must satisfy, together with the claim that providing a system satisfying that property appears difficult at humanity's present level of understanding, so our list of reasons translates fairly straightforwardly to a list of (defeasible) constraints on a suitable formalization of philosophy.

4.1 Difficulties arising from philosophical argumentation being broad

- it seems likely that a formal language in which philosophical arguments could be provided would imply a formal language in which any arguments could be provided, including eg arguments inside particular scientific theories, arguments that particular scientific theories are correct, arguments that a correct scientific theory would need to have such-and-such properties. so formalizing philosophy plausibly cannot be done without formalizing everything. the converse, that science (or just ordinary more descriptive reasoning) couldn't be formalized without philosophy, is also plausible. maybe it's clearer to ask for a formalization of everything
- maybe to get a meaningful chunk of philosophy, one really needs to effectively have a language in which all the informal reasoning that shows up in mathematics but hasn't been formalized yet could be formalized, because these informal questions can be asked as philosophical questions (to do: give examples). so maybe the distinction between all reasoning which goes into some conclusion and just statements and proofs/arguments doesn't help us (doesn't imply that our formalization project is more feasible) after all

4.2 The formal system would need to contain a lot of information about human concepts

- minimally, it seems likely that we will often need any compelling final argument to essentially deconfuse us about the question itself — it will sometimes need to restate a better version of the question. this is roughly because the deconfusion work cannot happen on the step where the problem is turned into a formal problem, because we don't want to offload this philosophical work to the system, not have to do it ourselves. so we need to be able to state confused questions in the formal language, and then provide arguments answering them; these arguments will probably have to involve stuff that looks like restating the problem, arguing about restatements, then solving it (insofar as anything remains to be solved at that point). in some sense, i think we are in extra trouble because 'a confused question cannot be translated — it only makes sense in the language in which it was asked' (meta-remark: but this obviously needs to be made more precise)
- examples of philosophical claims that sort of reference the entire language:
 - something with counterfactuals
 -
- this is speech to text so probably has errors: A lot of good philosophical work does not really have a concrete ethical aim in mind, at least not a single one. Instead, it takes a local patch of language, a collection of related terms, say, and seeks to reanalyze it fruitfully. in particular, seeks to reanalyze it so that it becomes more coherent, or becomes clearer, or relates more usefully to other parts of our understanding, allows progress to be made on other parts of our understanding. This kind of philosophical work seems really hard to formalize unless you have a language which captures all the relations of the language in which this kind of philosophical work would usually be carried out, i.e. human language. What do I mean by this language capturing all such relations? Well, I could roughly imagine two

ways in which this could happen. One is by taking this language at face value, having a language which extends this human natural language, which takes seriously the commitments made usually in human natural language, basically does the kind of reasoning a philosopher does when they take natural language, so-called intuitions, seriously, and they seek to reanalyze certain terms, to seek more precise definitions of certain terms, clearer notions which are in agreement with these natural language commitments involving the notions. A second way to do this is to, perhaps in the spirit of Rudolf Skarnap, think of these matters as pragmatic questions, allowing you to analyze them sort of from the outside of the language, from outside the language. So now you would be taking a look at a local patch, or an entirety I guess, of language and analyzing it from the outside, looking at it from a philosophical point of view. In this case, you needn't really take seriously the commitments made in this natural language, but you should probably know them, so that you can suggest reforms to the language which conform with existing common sense, so-called common sense or intuitive commitments in the language, to make it so that the reforms suggested are not too radical to be implemented easily. And perhaps also, simply keeping certain desirable properties of the existing language, properties which might not be explicit to you, sort of like Justin Zemschwet, there's a bunch we can do with language, but we don't really understand how we do it with language, it makes sense to keep, to stay true to the structure, to the commitment in existing language, to some degree, because these commitments let us do stuff and we don't really understand which properties we need, we'd have a hard time specifying explicitly, when looking at a language from the outside, the properties that are needed to do stuff in the language, we'd do a pretty good job if we, well the hope is we can do a pretty good job if we keep most of the properties it has, but reorganize it in minor ways. So there's these two different approaches to going about reforming our language, one sort of from the inside of the language, sort of like philosophy that takes intuition seriously, sort of the, I guess, Quinean version, and then there is a version that takes sort of an outside view on the language, we use the language from the outside, takes philosophical problems about, say, redefining terms to be pragmatic questions about constructing a fruitful language, a language in which thinking can be done truthfully, seeks to reform it from the outside. In either case, it seems that you need a language which knows quite a lot about this natural language. If you go with the former, you need to take natural language commitments seriously inside your language, if you go with the latter, you need to have a language which can talk about natural language commitments without taking them seriously necessarily, at least not on the meta level. Either way, you're going to need a lot of data inside your language that captures natural language, well, data in your reasoning system anyway, it could be inside your axioms, say, instead of being inside your terms and relations between terms.

maybe give free will example for refactoring (note that ethical commitments might be kept by the compatibilist notion)

- to add to the point about needing a particular language for the question to be stated: maybe related to the fact that philosophy proceeds more by amending previous understanding, dialectically, less by solving problems. solutions to problems are often (or at least are often coupled to) revisions to a previous conception that was causing some particular problem. For example, the philosophical problem of resolving Zeno's paradox of Achilles and the tortoise (https://en.wikipedia.org/wiki/Zeno%27s_paradoxes#Achilles_and_the_tortoise) might involve a reconceptualization (or clarification) of what motion is or of what it takes to complete tasks, saying something about why [the original counterargument to it being possible for Achilles to overtake the tortoise] does not go through in this conceptualization, and saying something about why the previous system of notions (in which the paradox was stated) was lacking. I find it difficult to imagine what it would look like to dissolve the paradox without being in meaningful contact with the confused conception in which it was stated. More precisely, one needs to be in contact with the confused conception in this case to be able to tell that one's reconceptualization indeed captures the original subject matter and the original argument, and to argue that it is an improvement on the confused conception.

4.3 The formal system would need to contain a lot of information about human purposes

- normativity: note immediately that if we had a language in which the question "should i do x?" can be appropriately stated and pertinent arguments can be provided, then it must immediately compel us
- more sophisticated point, with examples, on philosophical inquiry being purposive. when becoming less confused about what question is right to ask, we often check against some purposes that answering the present question is supposed to serve. these might not be explicit at all when starting with the original questions. examples:
 - Here's an example of a philosophical question: is everything that can be computed in our universe Turing computable? then can query further purposes if confusion arises: maybe care about figuring out what a Gödel sentence for science would look like
 - One might start the investigation in this paper from 'why can't we just formalize philosophy like we've formalized mathematics?'. One might then notice that one needs to clarify what kind of formalization is needed. To clarify what kind of formalization is needed, one considers to what end one wants a formalization, and realizes that one largely wanted a formalization for it to be used when creating a safe AGI. One now has some guidance for figuring out what kind of formalization one should be after — more precisely, that one should be after the kind which lets one build an AGI that safely performs formal reasoning in that formal philosophical system. If further

difficulties/uncertainties arise in understanding what kind of system is needed for that, then one might proceed to ask further questions about what one requires from this reasoning-setup in order to make the future be good.

- combining this point with the point in the previous section: insofar as purposes exist already, they are stated in our current informal language — this ties philosophical reasoning to our current informal language
- however, things might be worse still: maybe the ‘propositional content’ of assertions in current human language is insufficient to adequately pin down ethics. for instance, imagine someone experiencing some space thing (maybe pains from growing up in freefall, if that’s a thing), and then describing it in some hard-to-anticipate ethically relevant way, and then us empathizing and learning something ethical that wasn’t directly implicit in our linguistic data yet (but was perhaps directly implicit in our sum total understanding). something something moral experimentation, something something ethical attunement to the world (<https://joecarlsmith.com/2024/03/25/on-attunement>). This is distinct from whether it is possible for a team of alien scientists to deduce human values from linguistic data (data about what we say and write and assert, let’s say) — it seems quite plausible that the friendly scientist could do its job even if values are not already present in what we say in language in the direct sense — these alien scientists could do something quite different from trying to bootstrap ethical reflection from current human-language ethical assertions — for instance, they could perhaps instead determine the computationally relevant structure of an example human brain from this linguistic data and use the capacity to emulate its operation to extract a specification of human values.⁵

4.4 Potential desirable non-classical properties of the reasoning system

- despite being sorta a priori, sufficiently tied to the world for probabilistic/uncertain/defeasible argumentation to probably have to show up in any satisfactory formalization. give examples. talk about how people are working on this
- one issue with a defeasible reasoning system in particular being used in the context of an untrusted prover is that we might have lost all the safety, because, if it, say, wanted to convince us of something, it could plausibly provide lots of arguments in favor of that thing (<https://www.lesswrong.com/posts/34XxbRFe54Fyc0CDw/the-bottom-line>). Now, whether a particular system producing defeasible formal arguments will have this issue might depend on the system of defeasible reasoning as well as on how the argument-producing AI is trained/created/designed, though there are some general usual reasons from the AI alignment literature to worry — in any case, we will not go into the plausibility this in more detail here.
- the language might need to allow redefinitions of terms, and generally meaningful refactoring? might need to be more natively reflective? (ie can talk about itself) for one because it needs to reason about which redefinitions are good, which plausibly requires taking a meta-view. we might then just want to also have a metalanguage, but we might want to apply the same kind of scrutiny to the metalanguage. this quick proliferation, as well as pragmatic/meta- and object-considerations being hard to untangle makes it plausibly easier to try to have a single language where reflection is native

presently unclear if failing to have these properties in the formal system would at most cause merely pragmatic issues — i.e., if the formalization can be done in a classical system with at worst some loss in the formalization being easy to perform — or if serious — i.e., really need to find a suitable pretty non-classical system

5 What are some paths to formalizing philosophy?

- Try to develop a precise understanding of what it is to state a philosophical claim and to argue for it. in which philosophical reasoning some issues:
 - maybe there isn’t a very nice/principled formal structure to be found here? maybe one needs to embed a lot of messy stuff to get this right?
 - philosophers have been trying to do this for a while — this is what the philosophy of language and metaphilosophy, as well as some questions in the philosophy of mind, are largely about. a bunch of GOFAI (good old-fashioned AI) research also tried to do this. some neuroscience and psychology research can also be seen as attempting to do this. no one has succeeded yet, probably. that humans have tried a lot and not succeeded is an argument in favor of it being very difficult to develop this principled understanding of how philosophical reasoning ought to operate, indicating that it will perhaps be unlikely that we succeed at this before transformative AI is created using other methods.
- Try to mimic human thinking/reasoning very closely, possibly without developing much of an understanding of which aspects of human thinking/reasoning are important for our philosophical competence.

5. I admit this could be thorny, for one because what it is for the brain to e.g. reflect on philosophical matters appropriately is itself probably value-laden, so perhaps the alien might have some work set out for them in understanding what it is for them to emulate the operation of that human brain in a metaethically appropriate manner, but let me just claim that it seems plausible the alien could make something like this work.

6 Conclusions

- Philosophy seems hard to formalize. We’ve discussed various obstacles to its formalization: blabla
- In parallel, this provides a list of desiderata on a suitable formal system, constraining the space of possible designs.
- this is also all basically completely relevant to formalizing all of science
- it is also relevant to formalizing particular problems in science, or at least problems that relate to too many things for us to be able to cordon them off enough for a formalization which captures enough of the original problem to be available.
- while not a focus either, the obstacles are also obviously informative about what it would take for an ML-based attempt to produce a formalization of philosophy to succeed, and also informative to a lesser extent about what it takes for any attempt to get AI to do philosophy (or science) for us to succeed. but we will not make these connections precise here

References

- Dalrymple, David "davidad", Joar Skalse, Yoshua Bengio, Stuart Russell, Max Tegmark, Sanjit Seshia, Steve Omohundro, et al. 2024. *Towards Guaranteed Safe AI: A Framework for Ensuring Robust and Reliable AI Systems*. arXiv: 2405.06624 [cs.AI]. <https://arxiv.org/abs/2405.06624>.
- Po?lya, George. 1968. *Mathematics and Plausible Reasoning*. Princeton, N.J., Princeton University Press.
- Russell, Stuart. 2022. "Provably Beneficial Artificial Intelligence." In *Proceedings of the 27th International Conference on Intelligent User Interfaces*, 3. IUI '22. Helsinki, Finland: Association for Computing Machinery. ISBN: 9781450391443. <https://doi.org/10.1145/3490099.3519388>. <https://doi.org/10.1145/3490099.3519388>.
- Tegmark, Max, and Steve Omohundro. 2023. *Provably safe systems: the only path to controllable AGI*. arXiv: 2309.01933 [cs.CY]. <https://arxiv.org/abs/2309.01933>.